

# Smart Contract Audit Report

**Client:** ZeroDev

**Project:** Kernel v3.2 Incremental Audit

**Audit Period:** Nov 28, 2024 - Dec 4, 2024

**Auditors:** Felix Kim

## About the auditor

The journey began in 2019 at HAECHI LABS, where over 100 smart contracts were audited, focusing on Decentralized Finance (DeFi) systems.

Responsibilities included high-profile projects such as Shoyu and SuperRare, and the development of an automated auditing tool that enhanced both the efficiency and consistency of the auditing processes.

Continuing contributions to the field are made as a security researcher. Research is dedicated to enhancing security across various platforms and technologies. Projects include the development of an advanced fuzzing framework for detecting memory corruption vulnerabilities in Linux device drivers, and designing an emulation environment to detect security vulnerabilities in Android phone bootloaders through sophisticated fuzzing techniques. This depth of experience in practical security applications ensures a thorough and rigorous approach to smart contract audits.

# Audited Components

**Audit Base Commit:** 8c04088c67f40a37d0c924c3c1257125f10e1c77

**Audit Scope Latest Commit:** 0ff92c676addc1c9a7a183c1d48e87cf7fa97631

- **Replayable Signature**
  - **Description:** The replayable signature functionality is designed to be chain-agnostic, allowing the same signature to be used across multiple blockchain environments seamlessly.
- **Adopting solady's LibERC7579**
  - **Description:** Integration of Solady's LibERC7579 library to streamline the handling of ERC7579 modes and execution data.
- **Fix Minor Issues**
  - **Description:** Resolved unexpected edge cases in HookData handling to ensure consistent and reliable behavior during execution.
- **Code Refactoring**
  - **Description:** The calldata decoding logic has been simplified by introducing a structured format. Instead of manually calculating offsets and lengths using inline assembly, structs are used to represent the decoded calldata.

## Findings

No issues are found.

## **Disclaimer**

This audit report provides an independent evaluation of the smart contracts listed herein as requested by ZeroDev. The findings, interpretations, and conclusions expressed in this report are those of the auditors based on the information and source code made available to us at the time of the audit. This report does not constitute investment advice, legal advice, or any other type of advice.

While every effort has been made to ensure accuracy and thoroughness, the nature of software development and the risks associated with blockchain technology mean that absolute security cannot be guaranteed. This report should not be seen as an endorsement of any kind of the security practices involved. ZeroDev team is advised to continue conducting regular security assessments and updates to the audited contracts.